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Improving Autoregressive Conditional Diffusion for Fluid Flow Forecasting through Physics-Informed Losses and Multi-Step Prediction

by

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Abstract

Autoregressive conditional diffusion models show promise for fluid-flow forecasting, but remain limited by high inference cost and the lack of explicit physical constraints. This study targets physical consistency and inference efficiency by introducing flow-specific physics-informed regularisation, followed by multi-step conditional diffusion based on the selected physics-informed models. The models are evaluated on two-dimensional incompressible and transonic cylinder-flow benchmarks in terms of prediction accuracy, temporal behaviour, physical consistency, and inference efficiency. For incompressible flow, vorticity regularisation gave the strongest improvement, substantially reducing prediction and vorticity errors. Multi-step prediction reduced inference time but decreased accuracy, and additional temporal-consistency and horizon-weighting refinements did not recover this loss. For transonic flow, continuity regularisation improved the single-step model, while all tested multi-step horizons reduced prediction error and inference cost relative to the selected single-step reference, despite increased temporal-consistency errors and continuity residuals. Overall, the effects of physics-informed regularisation and multi-step prediction depend on the flow regime, the choice of physical constraint, and the multi-step prediction configuration. The results demonstrate trade-offs among prediction accuracy, temporal behaviour, physical consistency, and computational efficiency.

1 Introduction

Flow past a circular cylinder is a classical problem for studying viscous flow and wake-vortex evolution, but high-resolution numerical simulations of such flows are often computationally expensive. Data-driven surrogate models have therefore been widely explored as faster approximations to conventional numerical solvers by learning the spatiotemporal evolution of flow fields directly from simulation data [1, 2, 3].

This study builds on an autoregressive conditional diffusion model and extends it in two directions: physical consistency and inference efficiency. First, flow-specific physics-informed losses are introduced for different flow regimes to encourage greater agreement between the predicted fields and the governing physics. Second, single-step generation is extended to multi-step conditional diffusion, allowing multiple consecutive future states to be generated jointly within a single diffusion sampling process and thereby reducing the number of complete diffusion sampling calls required for long-horizon rollout. The proposed modifications are evaluated on both incompressible and transonic compressible cylinder-flow scenarios from four perspectives: prediction accuracy, temporal behaviour, physical consistency, and inference efficiency.

1.1 Problem Description and Significance

This study adopts an autoregressive conditional diffusion model as the baseline. The model predicts each subsequent flow state from a history of previous states and the corresponding physical parameters, and autoregressively reuses generated states as conditioning for subsequent predictions [2]. However, this formulation raises two main issues. First, the standard diffusion objective is primarily based on noise-prediction error and does not explicitly constrain the generated flow fields to satisfy the underlying physics. Consequently, a low prediction error does not necessarily imply good physical consistency [4]. Second, single-step autoregressive diffusion has a relatively high inference cost. Each future state requires a full reverse diffusion process, so the number of diffusion calls and denoising operations increases with rollout length [2].

To address these issues, this study introduces modifications in two directions: physical consistency and inference efficiency. Flow-specific physics-informed loss terms are incorporated to constrain relevant physical properties of the predicted fields. Building on the selected physics-informed configurations, single-step prediction is further extended to multi-step conditional diffusion, allowing multiple consecutive future states to be generated jointly within a single diffusion sampling process and thereby reducing

repeated sampling [5]. However, jointly predicting multiple future states also increases the dimensionality of the joint prediction target and may introduce an additional accuracy penalty.

These two directions are important for improving the practical value of diffusion-based flow surrogates. Greater physical consistency can improve the physical reliability of the predicted flow fields, while higher inference efficiency can reduce the computational cost of long rollouts and repeated forecasting tasks. However, these modifications may also affect other aspects of model performance: physics-informed regularisation may alter prediction accuracy, while multi-step generation may introduce accuracy or physical-consistency penalties despite improving efficiency. The proposed modifications are therefore evaluated from four complementary perspectives: prediction accuracy, temporal behaviour, physical consistency, and inference efficiency, allowing their benefits and trade-offs to be assessed together.

1.2 Review of Existing Work

1.2.1 Autoregressive Diffusion for Fluid Forecasting

Diffusion models learn complex data distributions through a gradual noising process followed by iterative denoising, and have subsequently been extended to PDE and fluid-flow prediction tasks [6, 7]. Kohl et al. [2] combined conditional diffusion with autoregressive rollout and proposed the Autoregressive Conditional Diffusion Model (ACDM), achieving competitive prediction accuracy and temporal stability across several two-dimensional flow benchmarks. However, in the single-step autoregressive formulation, each new physical state requires a complete reverse diffusion process, leading to substantially higher inference cost than conventional deterministic surrogate models [2]. In addition, long-horizon autoregressive forecasting remains susceptible to error accumulation, so model performance cannot be assessed from one-step error alone and should also consider rollout stability and temporal generalisation [8, 9, 10].

1.2.2 Physics-Informed Generative Modelling

To improve the physical consistency of generated flow fields, recent studies have incorporated governing physics into generative models. Physics-informed diffusion approaches can impose additional constraints through PDE residuals, physics-based loss terms, or sampling guidance [11, 12, 4]. Pi-fusion combines Navier–Stokes physics with diffusion-based prediction for velocity and pressure fields [4]. For incompressible flows, divergence-free diffusion models have further investigated different constraint strategies, including soft divergence penalties and hard projection methods, showing that soft regularisation can improve physical consistency but does not strictly guarantee conservation [13]. These studies demonstrate the potential of physics-informed regularisation for improving the physical reliability of generative flow prediction, while also suggesting that stronger physical constraints may introduce a trade-off with prediction quality.

1.2.3 Multi-Step and Multi-State Forecasting

Shysheya et al. [5] investigated a conditional diffusion formulation in which P future PDE states are generated conditioned on C preceding states, demonstrating that diffusion models can be extended from single-state prediction to joint multi-state generation. DiffFluid similarly adopts a temporal mapping from multiple historical states to multiple future states and demonstrates the ability of diffusion-based models to perform multi-step forecasting of complex fluid dynamics [3]. More broadly, studies of autoregressive PDE forecasting have shown that training horizon and rollout design can substantially affect long-term prediction performance [8, 9].

Overall, previous studies have explored autoregressive diffusion, physics-informed generative modelling, and multi-state forecasting from different perspectives. However, controlled studies that examine physics-informed regularisation and multi-step generation within a common ACDM framework, while jointly

evaluating their effects on prediction accuracy, temporal behaviour, physical consistency, and inference efficiency, remain relatively limited.

1.3 Objectives

The objectives of this project are:

1. To investigate flow-specific physics-informed loss terms for improving the physical consistency of diffusion-based flow forecasting and to assess their effect on prediction accuracy.
2. To investigate multi-step conditional diffusion as a means of reducing repeated diffusion sampling and inference cost.
3. To quantify the trade-offs among prediction accuracy, temporal behaviour, physical consistency, and computational efficiency across different prediction horizons.
4. To assess how the proposed modifications perform across incompressible and transonic flow regimes.

2 Methodology

2.1 Datasets and Flow Scenarios

This study uses two two-dimensional cylinder-flow datasets from the benchmark of Kohl et al. [14]: an incompressible wake case (Inc) generated with PhiFlow and a transonic compressible case (Tra) generated with SU2. The Inc state contains pressure p and velocity components u and v , with Reynolds number used as the conditioning parameter. The Tra state additionally includes density ρ , with Mach number used for conditioning. The Inc dataset contains 91 trajectories with 1300 snapshots each, while the Tra dataset contains 41 trajectories with 1001 snapshots each. Both datasets are represented on a 128×64 grid.

For evaluation, the Inc models are tested on held-out low- and high-Reynolds-number regimes, while the Tra models are evaluated on held-out Mach-number ranges following the benchmark split of Kohl et al. [2]. Detailed training and evaluation parameter ranges are summarised in Appendix B.

2.2 Baseline Autoregressive Conditional Diffusion Model

This study adopts the Autoregressive Conditional Diffusion Model (ACDM) of Kohl et al. [2] as the baseline. The model conditions on the two preceding flow states to predict the next state:

$$\mathbf{s}^t \sim f_{\theta} \left(\cdot \mid \mathbf{s}^{t-2}, \mathbf{s}^{t-1} \right). \quad (1)$$

During training, Gaussian noise is added to the target future state and the denoising network is trained to predict the added noise using a Huber loss:

$$\mathcal{L}_{\text{diff}} = \text{Huber}(\boldsymbol{\epsilon}, \boldsymbol{\epsilon}_{\theta}), \quad (2)$$

where $\boldsymbol{\epsilon}$ is the added noise and $\boldsymbol{\epsilon}_{\theta}$ is the noise predicted by the network. Noisy conditioning is also applied to the historical states, following Kohl et al. [2].

The reverse diffusion process is parameterised by a ConvNeXt-based U-Net with a three-level encoder–decoder structure, skip connections, attention modules, and timestep embeddings. The model uses 20 diffusion steps with a linear noise schedule. For Inc, the conditioning history contains p , u , and v together with the Reynolds number, while for Tra it contains p , ρ , u , and v together with the Mach number. Kohl et al. [2] report that increasing the number of diffusion steps beyond approximately 20 provides only limited

additional improvement for the Tra scenario, making this setting a reasonable compromise between predictive performance and inference cost.

During inference, each future state is generated through a complete reverse diffusion process and then fed back into the conditioning history:

$$\left(\mathbf{s}^{t-2}, \mathbf{s}^{t-1}\right) \rightarrow \hat{\mathbf{s}}^t, \quad \left(\mathbf{s}^{t-1}, \hat{\mathbf{s}}^t\right) \rightarrow \hat{\mathbf{s}}^{t+1}. \quad (3)$$

Thus, in single-step autoregressive forecasting, each new physical time step requires one complete diffusion sampling process.

2.3 Physics-Informed Loss

The standard diffusion objective is formulated in terms of noise prediction and therefore does not directly constrain the physical properties of the reconstructed flow fields. To introduce explicit physical regularisation, this study augments the diffusion objective with flow-specific physics-informed loss terms. Since the physical quantities of interest are defined on the flow fields themselves, these losses are evaluated on the reconstructed clean future states rather than directly on the predicted diffusion noise.

For the incompressible case (Inc), divergence and vorticity are used to constrain the predicted velocity field. An incompressible velocity field satisfies the divergence-free condition [1],

$$\nabla \cdot \mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0, \quad (4)$$

while the two-dimensional vorticity is defined as [1]

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \quad (5)$$

The divergence loss penalises the magnitude of the predicted divergence. The vorticity loss measures the mean-squared error between the predicted and ground-truth vorticity fields. The resulting objective is

$$\mathcal{L}_{\text{Inc}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{div}} \mathcal{L}_{\text{div}} + \lambda_{\text{vort}} \mathcal{L}_{\text{vort}}. \quad (6)$$

For the transonic compressible case (Tra), the incompressibility constraint is no longer applicable. Instead, mass conservation is imposed through the compressible continuity equation [1],

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} = 0. \quad (7)$$

The continuity loss is computed from the squared residual of this equation over the valid fluid region. The resulting training objective is

$$\mathcal{L}_{\text{Tra}} = \mathcal{L}_{\text{diff}} + \lambda_{\text{cont}} \mathcal{L}_{\text{cont}}. \quad (8)$$

The loss definitions, finite-difference evaluation, scaling, and masking procedures are summarised in Appendix A.

2.4 Multi-Step Conditional Diffusion

To reduce the repeated reverse-diffusion sampling required by the single-step baseline, this study extends ACDM to a multi-step formulation in which multiple consecutive future states are generated jointly within one diffusion process. Related conditional diffusion approaches have also explored joint generation of multiple future PDE states [5].

Let C denote the number of historical conditioning states and P the number of future states generated jointly. The single-step baseline uses $C = 2$ and $P = 1$, whereas the multi-step models use symmetric temporal windows with $C = P$:

$$\left(\mathbf{s}^{t-C}, \dots, \mathbf{s}^{t-1}\right) \rightarrow \left(\hat{\mathbf{s}}^t, \dots, \hat{\mathbf{s}}^{t+P-1}\right), \quad C = P. \quad (9)$$

During training, the P future states are concatenated along the channel dimension and treated as a joint diffusion target. The noisy-conditioning strategy, diffusion schedule, number of diffusion steps, and basic U-Net architecture are kept consistent with the single-step baseline, while the conditioning and target channel dimensions increase with C and P , respectively.

To investigate whether the Inc multi-step formulation could be further improved, two additional $C = P = 2$ refinements were evaluated separately. The first introduces a temporal-consistency loss,

$$\mathcal{L}_{\text{tem}} = \text{MSE}\left(\hat{\mathbf{s}}^{t+1} - \hat{\mathbf{s}}^t, \mathbf{s}^{t+1} - \mathbf{s}^t\right), \quad (10)$$

which encourages the predicted change between consecutive future states to match the corresponding ground-truth temporal increment.

The second refinement applies horizon-dependent weighting to the diffusion loss,

$$\mathcal{L}_{\text{diff}}^{\text{HW}} = w_1 \mathcal{L}_t + w_2 \mathcal{L}_{t+1}, \quad (11)$$

with $w_1 = 1.0$ and $w_2 = 0.5$, placing greater emphasis on the nearer prediction horizon. These two refinements are evaluated as separate variants rather than being combined into a single model.

For a rollout of T future states, the number of complete diffusion sampling calls is reduced from T in the single-step model to

$$\left\lceil \frac{T}{P} \right\rceil. \quad (12)$$

However, increasing P also increases the size of the conditioning and prediction tensors, so the wall-clock speed-up is not expected to scale exactly with P . The Inc experiments use $C = P = 2$, while the Tra experiments evaluate $C = P = 2, 3$, and 4. Detailed training settings and model configurations are summarised in Appendix B.

2.5 Evaluation Metrics

Model performance is evaluated from four complementary perspectives: prediction accuracy, temporal behaviour, physical consistency, and inference efficiency. The core metrics used in the main analysis are summarised in Table 1. A complete list of evaluation metrics for the Inc and Tra cases is provided in Appendix C.

Table 1: Core evaluation metrics used in the main experiments.

Category	Metrics	Purpose
Prediction accuracy	predMSE, relative L_2 , final-step MSE	Overall prediction error and long-horizon endpoint accuracy
Temporal behaviour	Inc: temporal consistency error; Tra: temporal increment MSE; rollout stability slope	Frame-to-frame consistency and error growth during rollout
Physical consistency	Inc: divergence MSE, vorticity MSE; Tra: continuity MAE and MSE	Incompressibility, vortical-structure accuracy, and mass conservation
Inference efficiency	Wall-clock inference time, diffusion sampling calls, reverse-denoising steps	Computational cost and accuracy-efficiency trade-off

3 Results

3.1 Effect of Physics-Informed Regularisation

Incompressible case. Table 2 shows that vorticity-only regularisation provided the best overall performance across both Reynolds-number regimes. With $\lambda_{\text{vort}} = 0.01$, predMSE decreased from 2.86×10^{-3} to 1.88×10^{-3} in the low-Reynolds-number regime and from 1.32×10^{-2} to 3.61×10^{-4} in the high-Reynolds-number regime. The same configuration also achieved the lowest vorticity MSE in both regimes and the lowest rollout stability slope in the high-Reynolds-number case. Divergence-only and combined regularisation did not provide the same overall improvement. The vorticity-only configuration was therefore selected for the subsequent Inc multi-step experiments.

Table 2: Single-step physics-informed ablation results for the Inc case. Lower values are better for all metrics.

Model	predMSE	rel. L_2	Stability slope	Div. MSE	Vort. MSE
<i>Low-Reynolds-number regime</i>					
Baseline	2.86e-3	0.183	6.79e-6	4.70e-6	3.17e-4
Div. 0.01	2.67e-3	0.176	1.28e-5	3.40e-6	2.94e-4
Vort. 0.01	1.88e-3	0.148	4.50e-6	4.37e-6	2.04e-4
Div. 0.01 + Vort. 0.01	3.04e-3	0.188	1.28e-5	4.06e-6	3.58e-4
Div. 0.003 + Vort. 0.003	1.98e-3	0.152	4.09e-6	5.00e-6	2.46e-4
Div. 0.003 + Vort. 0.01	8.93e-3	0.323	1.12e-5	7.24e-6	2.63e-4
<i>High-Reynolds-number regime</i>					
Baseline	1.32e-2	0.376	7.27e-5	2.13e-5	2.00e-3
Div. 0.01	6.22e-3	0.258	3.07e-5	2.07e-5	1.05e-3
Vort. 0.01	3.61e-4	0.062	3.88e-7	1.95e-5	7.98e-5
Div. 0.01 + Vort. 0.01	3.11e-3	0.182	1.74e-5	2.10e-5	6.32e-4
Div. 0.003 + Vort. 0.003	5.84e-3	0.250	3.44e-5	2.80e-5	1.19e-3
Div. 0.003 + Vort. 0.01	7.81e-3	0.289	1.56e-5	2.38e-5	4.18e-4

Transonic case. Table 3 shows that the stronger continuity regularisation with $\lambda_{\text{cont}} = 0.2$ provided the best overall performance. Relative to the single-step baseline, predMSE decreased from 2.93×10^{-2} to 2.69×10^{-2} , continuity MSE decreased from 3.05×10^{-4} to 2.03×10^{-4} , and the rollout stability slope decreased from 4.35×10^{-5} to 2.76×10^{-5} . In contrast, $\lambda_{\text{cont}} = 0.01$ did not improve prediction or continuity errors. The $\lambda_{\text{cont}} = 0.2$ configuration was therefore selected for the subsequent Tra multi-step experiments.

Table 3: Single-step continuity-loss results for the Tra case. Lower values are better for all metrics.

Model	predMSE	rel. L_2	Stability slope	Cont. MAE	Cont. MSE
Baseline	2.93e-2	0.266	4.35e-5	1.03e-2	3.05e-4
Cont. 0.01	3.26e-2	0.281	2.84e-5	1.10e-2	3.85e-4
Cont. 0.2	2.69e-2	0.255	2.76e-5	9.31e-3	2.03e-4

3.2 Multi-Step Prediction Accuracy

Incompressible case. Table 4 shows that the vorticity-regularised single-step model remained the most accurate Inc configuration across both Reynolds-number regimes. Extending it to $C = P = 2$ increased predMSE from 1.88×10^{-3} to 5.51×10^{-3} in the low-Reynolds-number regime and from 3.61×10^{-4} to 1.93×10^{-3} in the high-Reynolds-number regime. Relative L_2 and final-step MSE showed the same deterioration.

The two additional $C = P = 2$ refinements, both retaining the selected vorticity regularisation, also failed to recover this loss in accuracy. Both the temporal-consistency and horizon-weighting variants produced substantially higher predMSE than the basic multi-step model in both regimes. Among the tested Inc multi-step variants, the basic vorticity-regularised $C = P = 2$ model therefore achieved the best prediction accuracy.

Table 4: Prediction accuracy of the Inc single-step and multi-step models over 500-step rollouts. Lower values are better for all metrics.

Model	predMSE	rel. L_2	Final-step MSE
<i>Low-Reynolds-number regime</i>			
SS baseline	2.86e-3	0.183	4.42e-3
SS + vort.	1.88e-3	0.148	3.23e-3
MS ($C = P = 2$) + vort.	5.51e-3	0.253	6.13e-3
MS ($C = P = 2$) + vort. + temp.	2.42e-2	0.531	1.29e-2
MS ($C = P = 2$) + vort. + HW	1.62e-2	0.434	1.06e-2
<i>High-Reynolds-number regime</i>			
SS baseline	1.32e-2	0.376	3.21e-2
SS + vort.	3.61e-4	0.062	4.69e-4
MS ($C = P = 2$) + vort.	1.93e-3	0.143	2.27e-3
MS ($C = P = 2$) + vort. + temp.	3.34e-2	0.597	6.21e-2
MS ($C = P = 2$) + vort. + HW	2.95e-2	0.561	6.18e-2

Transonic case. Table 5 shows a different trend for Tra: all tested multi-step models outperformed the continuity-regularised single-step reference in overall prediction accuracy. The $C = P = 2$ model achieved the lowest predMSE, reducing it from 2.69×10^{-2} to 2.16×10^{-2} , while relative L_2 decreased from 0.255 to 0.228.

Increasing the joint horizon to $C = P = 3$ and $C = P = 4$ caused only a small increase in average prediction error, with predMSE values of 2.17×10^{-2} and 2.21×10^{-2} , respectively. In contrast, final-step MSE improved progressively with horizon, reaching its lowest value of 2.36×10^{-2} for $C = P = 4$. Thus, $C = P = 2$ achieved the best overall accuracy, while $C = P = 4$ achieved the lowest final-step error.

3.3 Temporal and Physical Metrics

Incompressible case. Table 6 shows that vorticity regularisation improved both temporal and physical metrics for the single-step Inc model. In the low-Reynolds-number regime, temporal consistency error decreased from 8.34×10^{-5} to 5.92×10^{-5} and vorticity MSE from 3.17×10^{-4} to 2.04×10^{-4} . The

Table 5: Prediction accuracy of the Tra single-step and multi-step models over 500-step rollouts. Lower values are better for all metrics.

Model	predMSE	rel. L_2	Final-step MSE
SS baseline	2.93e-2	0.266	3.37e-2
SS + cont.	2.69e-2	0.255	2.69e-2
MS ($C = P = 2$) + cont.	2.16e-2	0.228	2.55e-2
MS ($C = P = 3$) + cont.	2.17e-2	0.229	2.42e-2
MS ($C = P = 4$) + cont.	2.21e-2	0.231	2.36e-2

improvement was stronger in the high-Reynolds-number regime, where the corresponding values decreased to 4.34×10^{-5} and 7.98×10^{-5} , respectively.

The basic $C = P = 2$ multi-step model achieved the lowest rollout stability slope in both regimes, but its temporal and physical errors were higher than those of the vorticity-regularised single-step model. The corresponding rollout-error curves in Appendix D further show that the lower stability slope of the basic multi-step model reflects slower error growth rather than lower absolute prediction error. The additional temporal-consistency loss further increased temporal, divergence, and vorticity errors and did not improve the intended temporal behaviour.

Table 6: Temporal and physical metrics for representative Inc models over 500-step rollouts. Lower values are better.

Model	Temp. consistency error	Stability slope	Div. MSE	Vort. MSE
<i>Low-Reynolds-number regime</i>				
SS baseline	8.34e-5	6.79e-6	4.70e-6	3.17e-4
SS + vort.	5.92e-5	4.50e-6	4.37e-6	2.04e-4
MS ($C = P = 2$) + vort.	2.22e-4	1.42e-6	3.11e-5	9.18e-4
MS ($C = P = 2$) + vort. + temp.	6.64e-4	2.20e-5	4.26e-5	2.07e-3
<i>High-Reynolds-number regime</i>				
SS baseline	5.35e-4	7.27e-5	2.13e-5	2.00e-3
SS + vort.	4.34e-5	3.88e-7	1.95e-5	7.98e-5
MS ($C = P = 2$) + vort.	9.71e-5	6.41e-8	2.38e-5	1.27e-4
MS ($C = P = 2$) + vort. + temp.	1.21e-3	1.46e-4	5.33e-5	3.87e-3

Transonic case. Table 7 shows that the continuity-regularised single-step model reduced continuity MSE from 3.05×10^{-4} to 2.03×10^{-4} and the rollout stability slope from 4.35×10^{-5} to 2.76×10^{-5} .

Table 7: Temporal and physical metrics for the Tra single-step and multi-step models over 500-step rollouts. Lower values are better.

Model	Temp. inc. MSE	Stability slope	Cont. MAE	Cont. MSE
SS baseline	1.48e-3	4.35e-5	1.03e-2	3.05e-4
SS + cont.	1.67e-3	2.76e-5	9.31e-3	2.03e-4
MS ($C = P = 2$) + cont.	2.44e-3	2.61e-5	4.24e-2	2.78e-3
MS ($C = P = 3$) + cont.	3.83e-3	1.75e-5	5.44e-2	4.64e-3
MS ($C = P = 4$) + cont.	2.93e-3	1.56e-5	5.41e-2	4.64e-3

The multi-step models further reduced the stability slope as the joint horizon increased, reaching 1.56×10^{-5} for $C = P = 4$. The rollout-error curves in Appendix D similarly show that the multi-step models generally maintain lower prediction errors than the single-step continuity reference over much of the rollout. However, all multi-step variants showed larger temporal increment errors and substantially larger continuity residuals than the single-step models. Continuity MSE increased to 2.78×10^{-3} for $C = P = 2$ and to approximately

4.64×10^{-3} for $C = P = 3$ and $C = P = 4$. Thus, the improved rollout accuracy reported in Section 3.2 was accompanied by weaker local temporal and mass-conservation consistency. The pressure-field and absolute-error maps in Appendix D show the spatial distribution of the pressure errors around the predicted wake structures. Although the multi-step models achieve lower aggregate prediction errors, the field visualisations indicate that lower MSE does not necessarily correspond to sharper or more faithful local flow structures, particularly for the longer $C = P = 4$ horizon.

3.4 Inference Efficiency

The reported sampling counts are aggregated across five Inc rollouts and two Tra rollouts, each containing 500 predicted states.

Incompressible case. Table 8 shows that the $C = P = 2$ formulation approximately halved the diffusion sampling cost relative to the vorticity-regularised single-step model. The number of diffusion calls decreased from 2500 to 1250 and the number of reverse denoising steps from 50000 to 25000. This produced a wall-clock speed-up of $1.99\times$ in the low-Reynolds-number regime and $1.97\times$ in the high-Reynolds-number regime.

Table 8: Inference efficiency of representative Inc models over 500-step rollouts. Speed-up is measured relative to the vorticity-regularised single-step model.

Model	Time (min)	Diff. calls	Reverse steps	Speed-up	predMSE
<i>Low-Reynolds-number regime</i>					
SS + vort.	8.70	2500	50000	1.00×	1.88e-3
MS ($C = P = 2$) + vort.	4.37	1250	25000	1.99×	5.51e-3
MS ($C = P = 2$) + vort. + temp.	4.40	1250	25000	1.98×	2.42e-2
<i>High-Reynolds-number regime</i>					
SS + vort.	8.25	2500	50000	1.00×	3.61e-4
MS ($C = P = 2$) + vort.	4.19	1250	25000	1.97×	1.93e-3
MS ($C = P = 2$) + vort. + temp.	4.19	1250	25000	1.97×	3.34e-2

The temporal-loss variant had essentially the same inference cost as the basic $C = P = 2$ model because both used the same prediction horizon. As shown in Section 3.2, however, the efficiency gain of the Inc multi-step models was accompanied by higher prediction error than the vorticity-regularised single-step reference.

Transonic case. Table 9 shows that increasing the joint prediction horizon progressively reduced the inference cost of the Tra models. Relative to the continuity-regularised single-step reference, the measured speed-up increased from $1.87\times$ for $C = P = 2$ to $2.32\times$ for $C = P = 3$ and $2.79\times$ for $C = P = 4$, with inference time decreasing to 1.82, 1.47, and 1.22 min, respectively.

Table 9: Inference efficiency of the Tra continuity-regularised single-step and multi-step models over 500-step rollouts. Speed-up is measured relative to the single-step continuity model.

Model	Time (min)	Diff. calls	Reverse steps	Speed-up	predMSE
SS + cont.	3.41	1000	20000	1.00×	2.69e-2
MS ($C = P = 2$) + cont.	1.82	500	10000	1.87×	2.16e-2
MS ($C = P = 3$) + cont.	1.47	334	6680	2.32×	2.17e-2
MS ($C = P = 4$) + cont.	1.22	250	5000	2.79×	2.21e-2

Figure 1 shows the corresponding accuracy–efficiency trade-off. The $C = P = 2$ model achieved the lowest predMSE, while $C = P = 4$ achieved the shortest inference time. Increasing the horizon therefore improved efficiency with only a modest increase in prediction error among the tested multi-step models.

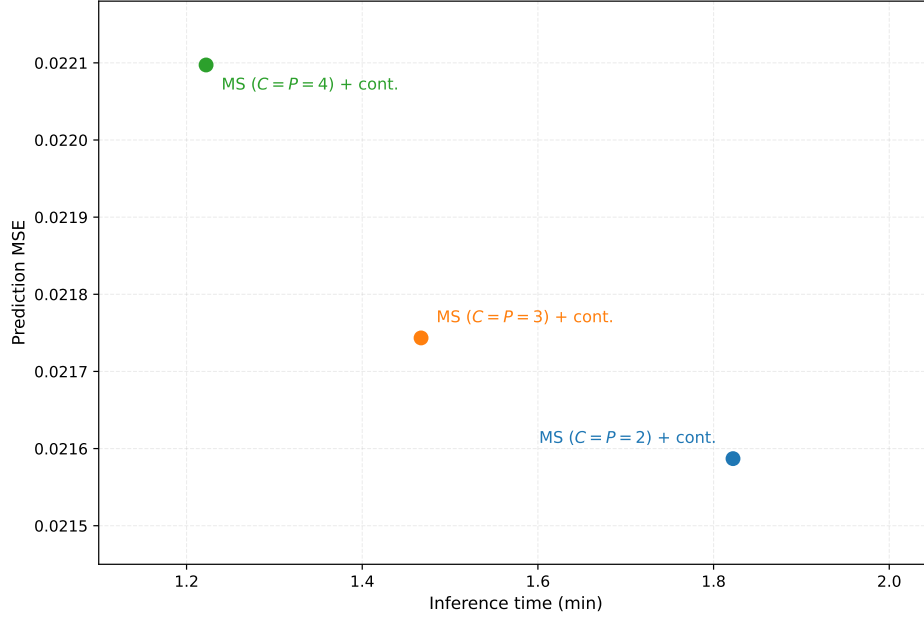


Figure 1: Accuracy–efficiency trade-off among the Tra multi-step models. Lower-left points indicate lower prediction error and shorter inference time. All models use continuity regularisation.

4 Discussion

4.1 Effectiveness of Physics-Informed Regularisation

The Inc results show that introducing additional physical constraints does not necessarily lead to better predictive performance. Across both Reynolds-number test regimes, vorticity-only regularisation achieved the best overall prediction accuracy while also substantially reducing the vorticity error. One possible explanation is that the evolution of the cylinder wake is strongly associated with periodic vortex shedding and coherent vortical structures, as extensively discussed in classical studies of cylinder wakes [15, 16]. The strong performance of vorticity regularisation is therefore consistent with directly constraining a flow quantity closely related to these wake structures, although the present experiments do not isolate this mechanism directly. By contrast, the divergence loss primarily constrains incompressibility through the divergence-free condition [1], and a lower divergence residual does not necessarily correspond to a lower overall field-prediction error. The Inc weight experiments also did not show a simple relationship in which stronger regularisation consistently improved performance. Reducing the combined-loss weight from 0.01 to 0.003 improved some prediction metrics, but neither combined configuration outperformed the vorticity-only model. Because the physical terms are introduced as penalty losses, they act as soft constraints: they encourage the model to satisfy the corresponding physical conditions but do not guarantee exact enforcement. Related work similarly distinguishes penalty-based soft enforcement from projection-based hard enforcement and notes that soft constraints generally provide only approximate physical consistency rather than exact conservation [13].

The Tra results exhibit a different sensitivity to the continuity-loss weight. Among the two tested settings, $\lambda_{\text{cont}} = 0.01$ did not improve either prediction error or continuity residuals, whereas $\lambda_{\text{cont}} = 0.2$ reduced both. Thus, within the tested range, the stronger continuity weight provided a better balance between prediction accuracy and mass-conservation consistency, although the present experiments are insufficient to identify a universally optimal weight or threshold. The continuity equation describes local mass conservation in compressible flow [1], but the ground-truth fields themselves exhibit a non-zero residual under the present discrete continuity metric. This indicates that the residual contains contributions from discretisation and post-processing and should therefore be interpreted primarily as a relative consistency metric rather than as an exact dimensional measure of mass-conservation error. In other words, the

continuity metric is used here to compare models under the same discrete diagnostic rather than to demonstrate exact satisfaction of the governing equation. Related physics-informed generative modelling studies likewise treat residual- or penalty-based constraints as soft consistency mechanisms rather than strict guarantees of conservation [12, 13].

Overall, the results indicate that the effectiveness of physics-informed regularisation depends on the selected physical quantity, regularisation weight, and flow regime, rather than increasing automatically with the number or strength of imposed constraints. Previous physics-informed diffusion studies have incorporated physical information through several different mechanisms, including PDE residuals, physics-based loss terms, and physics-guided sampling [4, 11, 12]. These approaches are not necessarily equivalent. For example, the ablation study in Pi-fusion shows that its PDE-loss variant performs substantially worse than physics-guided sampling, demonstrating that the inclusion of physical information alone does not guarantee improved prediction quality [4]. The present results are consistent with this observation: vorticity-only regularisation was more effective than divergence-only or combined regularisation for Inc, while continuity regularisation for Tra produced a clear benefit only for the larger of the two tested weights. Physical constraints and their weights should therefore be selected according to the specific forecasting task rather than treated as universally or monotonically beneficial modifications.

4.2 Accuracy–Efficiency Trade-off of Multi-Step Prediction

The efficiency results in this study confirm the direct benefit of the multi-step formulation. For Inc, the $C = P = 2$ model halved both the number of diffusion calls and reverse-denoising steps and achieved an almost $2\times$ wall-clock speed-up. For Tra, increasing the joint prediction horizon from $P = 2$ to $P = 4$ increased the measured speed-up from approximately $1.87\times$ to $2.79\times$. However, the observed acceleration remained below the ideal P -fold scaling. Although fewer diffusion calls are required, a larger joint target also increases the channel dimensions of the conditioning and prediction tensors, making each U-Net evaluation more expensive. The efficiency gain of multi-step prediction therefore depends on the balance between the increased cost of each sampling call and the reduction in the total number of calls. Related diffusion-based sequence forecasting methods have likewise explored multi-step, variable-horizon, or full-sequence generation to address limitations of purely step-by-step temporal prediction [17, 18].

The effect of multi-step prediction on forecasting accuracy differed substantially between the two flow scenarios. For Inc, the $C = P = 2$ formulation reduced inference cost but introduced a clear accuracy penalty relative to the vorticity-regularised single-step reference, resulting in a direct accuracy–efficiency trade-off. In contrast, all tested Tra multi-step configurations achieved both shorter inference times and lower predMSE than the continuity-regularised single-step reference. Among these models, $P = 2$ achieved the lowest overall prediction error, whereas $P = 4$ achieved the highest inference efficiency. These results show that multi-step generation does not necessarily improve efficiency at the expense of prediction accuracy; its effect depends on the particular flow regime and multi-step configuration.

One possible explanation is related to the reduction in autoregressive feedback. In autoregressive forecasting, previously generated states are repeatedly reused as conditioning inputs for subsequent predictions. As a result, early prediction errors can propagate forward and may gradually shift the inputs away from the ground-truth distribution encountered during training. Previous work on autoregressive PDE surrogates has shown that repeatedly using model predictions as future inputs allows early prediction errors to influence subsequent states throughout the rollout [19]. Similar issues also arise in learned physical simulators, where training strategies such as noise injection have been used to improve robustness to model-generated rollout states [20]. Solver-in-the-loop and autoregressive PDE benchmark studies have further emphasised the importance of recurrent deployment, unrolled training, and long-horizon evaluation for stable forecasting [8, 9]. Because multi-step generation produces several future states within a single diffusion process, it reduces how often the model reuses its own predictions as conditioning inputs and may therefore reduce repeated autoregressive error propagation.

On the other hand, a larger joint prediction target may also increase modelling difficulty because multiple

future states must be represented simultaneously within the same diffusion process. Multi-step forecasting can therefore be viewed as involving two competing effects: reduced autoregressive feedback and increased joint-prediction difficulty. The lower prediction errors of the Tra multi-step models are consistent with a setting in which the first effect is more beneficial, whereas the Inc results exhibit a more pronounced joint-prediction accuracy penalty. However, the present experiments do not isolate these two factors, so this interpretation should be regarded as a possible mechanism rather than a directly established causal explanation.

Overall, the preferred multi-step configuration differs between the two tested flow scenarios. For Inc, $C = P = 2$ provides an almost twofold inference speed-up but at the cost of a clear reduction in prediction accuracy. For Tra, increasing the joint prediction horizon produces a more favourable accuracy–efficiency relationship. At the same time, the results in Section 3.3 show that lower prediction error and higher inference efficiency do not necessarily correspond to improved physical consistency, particularly because the continuity residuals of the Tra multi-step models increase substantially. The prediction horizon should therefore be selected by considering prediction accuracy, temporal behaviour, physical consistency, and computational efficiency jointly.

4.3 Differences Between Flow Regimes

The Inc and Tra cases responded differently to multi-step conditional diffusion. In Inc, the vorticity-regularised single-step model already achieved a relatively low prediction error, whereas the $C = P = 2$ multi-step formulation substantially improved inference efficiency but introduced a clear accuracy penalty. By contrast, all tested Tra multi-step models achieved lower predMSE than the continuity-regularised single-step reference while also reducing inference time. Kohl et al. likewise showed that autoregressive diffusion models can exhibit different prediction accuracy, temporal stability, and long-horizon behaviour across different flow scenarios [2]. The present results therefore further indicate that the effect of multi-step generation cannot be assessed from prediction horizon alone, but should be considered together with the specific flow regime and the performance of the corresponding single-step reference.

For Inc, the relationship between cylinder-wake dynamics and vorticity regularisation has already been discussed in Section 4.1. Of greater relevance here is that, once the vorticity-regularised single-step reference had already reached a relatively low prediction error, the $C = P = 2$ model mainly provided an inference-efficiency benefit rather than a further reduction in prediction error. As discussed in Section 4.2, the balance between reduced autoregressive feedback and increased joint-prediction difficulty provides one possible explanation for this behaviour, although the present experiments do not isolate these effects.

The Tra case showed a more favourable response to multi-step generation. This scenario involves transonic compressible cylinder flow, in which pressure, density, and velocity are coupled through the governing conservation laws of compressible flow [1]. Previous numerical studies of transonic cylinder flow have also shown that compressibility can substantially alter the pressure distribution and downstream wake structure around the cylinder [21]. Unlike Inc, the Tra multi-step models reduced overall prediction error while also requiring fewer sampling calls. This result is also consistent with the possible benefit of reduced autoregressive feedback discussed in Section 4.2, although the present experiments do not establish this mechanism directly.

The two regimes also differ in the relationship between prediction accuracy and physical consistency. For Inc, the selected vorticity regularisation improved both prediction accuracy and vorticity consistency in the single-step setting, as discussed in Section 4.1. For Tra, the continuity residual measures local mass-conservation consistency, whereas predMSE measures the overall fieldwise prediction error across pressure, density, and velocity; these two quantities therefore evaluate different aspects of model performance [1]. A lower predMSE does not necessarily imply a lower continuity residual. The field visualisations further suggest that lower aggregate error does not necessarily imply better preservation of local flow structures, particularly for the longer multi-step horizon. This is reflected in the Tra multi-step results, where prediction accuracy and inference efficiency improved relative to the single-step reference while

continuity residuals increased. The preferred model for a given flow regime therefore cannot be determined from a single accuracy metric alone.

Overall, the two tested flow regimes do not share a single preferred multi-step configuration. For Inc, the vorticity-regularised single-step model remains preferable when prediction accuracy and physical consistency are prioritised, whereas the $C = P = 2$ multi-step model provides a more efficiency-oriented alternative. For Tra, the tested multi-step horizons provide a more favourable accuracy–efficiency relationship, although the deterioration in continuity consistency shows that these gains do not extend to all aspects of model performance. Multi-step prediction and horizon selection should therefore be evaluated in relation to the underlying flow dynamics, the performance of the single-step reference, autoregressive rollout behaviour, and the physical constraints relevant to the forecasting task, rather than assuming that the same multi-step strategy would have equivalent effects across different flow regimes.

4.4 Limitations and Future Work

This study has both methodological and experimental limitations.

From a methodological perspective, the physics-informed terms act as soft constraints and therefore do not strictly enforce incompressibility or mass conservation. In addition, some physical residuals were computed using discrete finite differences with normalised spatial and temporal intervals, so they are more suitable for relative model comparison than as exact dimensional residuals.

From an experimental perspective, the tested multi-step horizons were limited. For Inc, longer horizons were not explored because $C = P = 2$ already introduced a clear accuracy penalty, while Tra was evaluated only up to $C = P = 4$. The conclusions therefore apply only to the tested range. Physics-loss weights were also selected through limited sensitivity tests rather than systematic optimisation or adaptive weighting, and the chosen values should not be interpreted as globally optimal. The experiments were further restricted to cylinder-flow benchmarks and individual trained models.

Future work should therefore consider broader flow regimes and geometries, repeated runs with multiple random seeds, more systematic physics-loss weighting, and improved temporal or physical constraints for multi-step prediction.

5 Conclusions

This study investigated two extensions to an autoregressive conditional diffusion model for two-dimensional cylinder-flow forecasting: flow-specific physics-informed regularisation and multi-step conditional diffusion. The proposed models were evaluated on incompressible and transonic flow regimes in terms of prediction accuracy, temporal behaviour, physical consistency, and inference efficiency.

For the incompressible case, vorticity regularisation provided the strongest overall improvement, substantially reducing prediction and vorticity errors. However, extending the model to $C = P = 2$ multi-step prediction introduced a clear accuracy penalty, although inference time was reduced by approximately a factor of two. The additional temporal-consistency and horizon-weighting refinements did not recover this loss in accuracy.

For the transonic case, continuity regularisation improved both prediction accuracy and continuity consistency in the single-step model. Multi-step prediction produced a more favourable accuracy–efficiency trade-off: all tested horizons reduced prediction error relative to the selected single-step reference while also lowering inference cost. However, the multi-step models exhibited larger temporal and continuity residuals, showing that improved prediction accuracy and efficiency do not necessarily imply improved physical consistency.

Overall, the study achieved its objectives of evaluating flow-specific physics-informed regularisation, assessing multi-step conditional diffusion as a means of reducing repeated sampling cost, quantifying the

resulting trade-offs among prediction accuracy, temporal behaviour, physical consistency, and inference efficiency, and comparing these effects across incompressible and transonic flow regimes. The results show that no single configuration is uniformly preferable across all evaluation dimensions. Instead, the choice of physical regularisation and multi-step prediction horizon should depend on the flow regime and the relative importance of prediction accuracy, temporal behaviour, physical consistency, and computational efficiency.

Appendix

A Physics-Loss Implementation Details

Table 10: Physics-informed loss terms used for the two flow scenarios.

Case	Loss	Definition
Inc	Divergence	$\mathcal{L}_{\text{div}} = \text{mean}(\nabla \cdot \hat{\mathbf{u}})$
Inc	Vorticity	$\mathcal{L}_{\text{vort}} = \text{MSE}(\hat{\omega}, \omega)$
Tra	Continuity	$\mathcal{L}_{\text{cont}} = \text{mean}(R_{\text{cont}}^2)$

For the Inc losses, spatial derivatives are evaluated directly on the discrete computational grid. For the Tra continuity loss, u , v , and ρ are first transformed back to the original data scale. Temporal derivatives are evaluated between consecutive states, midpoint values are used for the mass fluxes, and spatial derivatives are computed using central finite differences. The solid obstacle, neighbouring cells, and outer grid boundary are excluded from the residual evaluation.

B Experimental Configuration

Table 11: Common training configuration used across the experiments.

Setting	Value
Training epochs	70
Batch size	2
Learning rate	10^{-4}
Diffusion steps	20
Noise schedule	Linear
Denoising backbone	ConvNeXt-based U-Net

Table 12: Summary of the experimental design for the two flow scenarios.

Case	Training / model variants	Evaluation ranges
Inc	Single-step physics-loss ablation followed by $C = P = 2$ multi-step experiments using the selected physics configuration. Separate temporal-consistency and horizon-weighting refinements were also evaluated.	Held-out $Re = 100-200$ and $Re = 900-1000$.
Tra	Single-step continuity-loss comparison followed by multi-step models with $C = P = 2, 3, 4$ using the selected continuity configuration.	Training: $Ma = 0.53-0.63$ and $0.69-0.90$. Evaluation: $Ma = 0.50-0.52, 0.64-0.65$, and $0.66-0.68$ [2].

C Evaluation Metrics

Table 13: Evaluation metrics for the Inc case.

Category	Metric	Purpose
Accuracy	predMSE	Overall rollout prediction accuracy
	reL2	Scale-normalised overall field error
	MSE_u	Horizontal velocity prediction accuracy
	MSE_v	Vertical velocity prediction accuracy
	MSE_pressure	Pressure prediction accuracy
	reL2_u	Scale-normalised horizontal velocity error
	reL2_v	Scale-normalised vertical velocity error
	reL2_pressure	Scale-normalised pressure error
Temporal behaviour	temporalContinuityError	Frame-to-frame temporal consistency
	MSE per timestep	Error evolution over rollout
	reL2 per timestep	Normalised error evolution over rollout
	rolloutStabilitySlope	Rate of long-horizon error accumulation
	final-step MSE	Long-horizon endpoint accuracy
Mass physics	predDivMean	Mean absolute divergence of predicted velocity field
	predDivMSE	Mean squared divergence of predicted velocity field
	gtDivMean	Reference mean absolute divergence of ground truth
	gtDivMSE	Reference mean squared divergence of ground truth
Flow structure	vorticityMSE	Vorticity / wake-structure accuracy
	vorticityReL2	Normalised vorticity-structure error
Efficiency	inferenceTimeMin	Total inference runtime in minutes
	numDiffusionCalls	Number of autoregressive diffusion sampling calls
	numReverseDenoisingSteps	Total reverse-denoising workload

Table 14: Evaluation metrics for the Tra case.

Category	Metric	Purpose
Accuracy	predMSE	Overall rollout prediction accuracy
	reL2	Scale-normalised overall field error
	MSE_u	Horizontal velocity prediction accuracy
	MSE_v	Vertical velocity prediction accuracy
	MSE_density	Density prediction accuracy
	MSE_pressure	Pressure prediction accuracy
	reL2_u	Scale-normalised horizontal velocity error
	reL2_v	Scale-normalised vertical velocity error
	reL2_density	Scale-normalised density error
	reL2_pressure	Scale-normalised pressure error
Temporal behaviour	temporalIncrementMSE	Error in the predicted frame-to-frame temporal increment
	MSE per timestep	Error evolution over rollout
	reL2 per timestep	Normalised error evolution over rollout
	rolloutStabilitySlope	Rate of long-horizon error accumulation
	final-step MSE	Long-horizon endpoint accuracy
Mass physics	predContinuityMAE	Mean absolute continuity residual of prediction
	predContinuityMSE	Mean squared continuity residual of prediction
	gtContinuityMAE	Reference continuity-residual MAE of ground truth
	gtContinuityMSE	Reference continuity-residual MSE of ground truth
Flow structure	vorticityMSE	Vorticity / rotational-structure accuracy
	vorticityReL2	Normalised vorticity-structure error
Transonic structure	pressureGradientMSE	Pressure-gradient structure accuracy
	pressureGradientReL2	Normalised pressure-gradient error
	densityGradientMSE	Density-gradient structure accuracy
	densityGradientReL2	Normalised density-gradient error
Efficiency	inferenceTimeMin	Total inference runtime in minutes
	numDiffusionCalls	Number of autoregressive diffusion sampling calls
	numReverseDenoisingSteps	Total reverse-denoising workload

D Additional Experimental Results

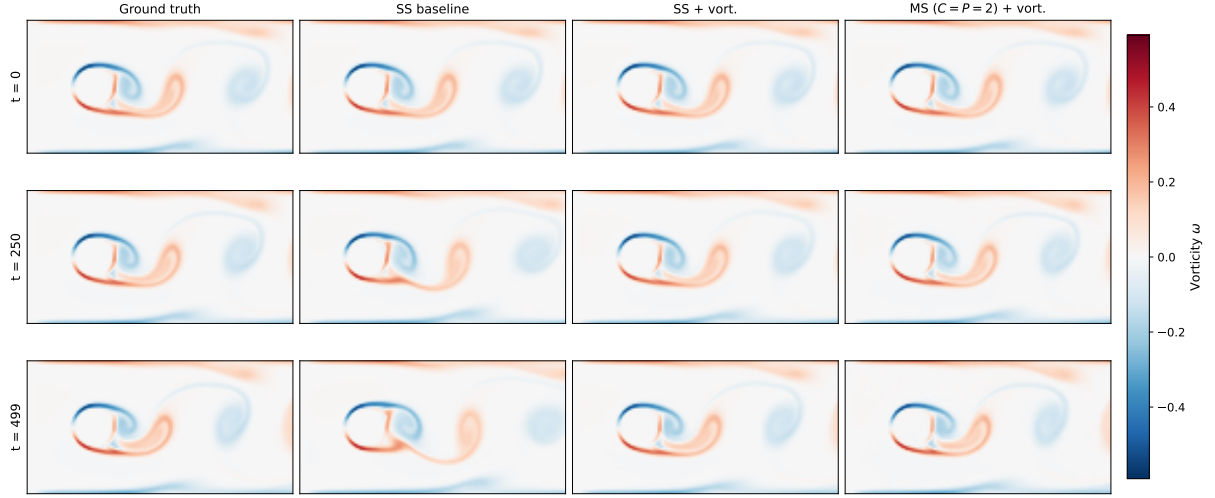


Figure 2: Vorticity-field comparison for the high-Reynolds-number Inc case at rollout indices $t = 0$, 250, and 499. Columns show the ground truth, single-step baseline, vorticity-regularised single-step model, and vorticity-regularised $C = P = 2$ multi-step model.

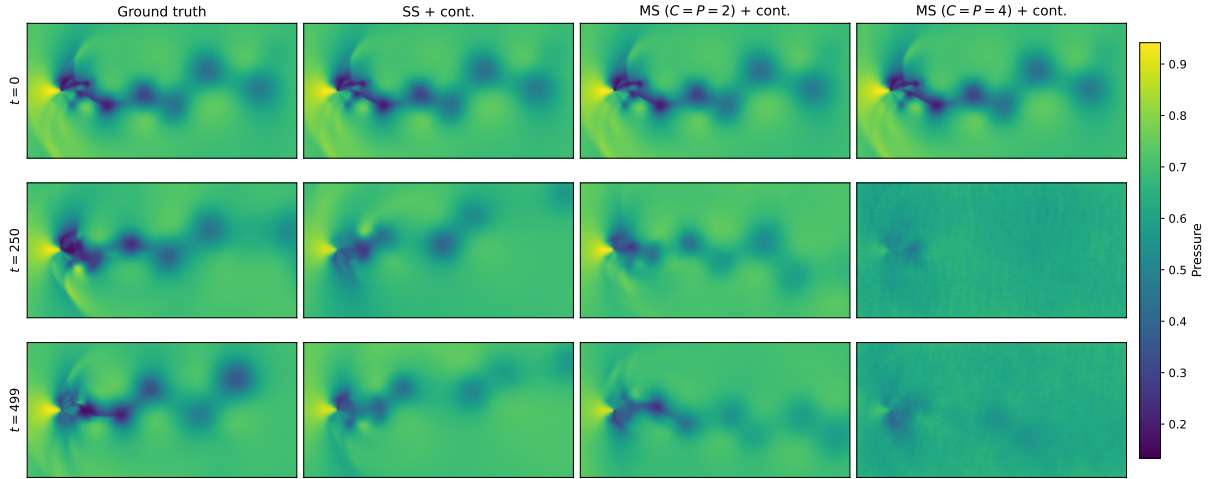


Figure 3: Pressure-field comparison for the Tra case at rollout indices $t = 0$, 250, and 499. Columns show the ground truth, continuity-regularised single-step model, and the $C = P = 2$ and $C = P = 4$ multi-step models.

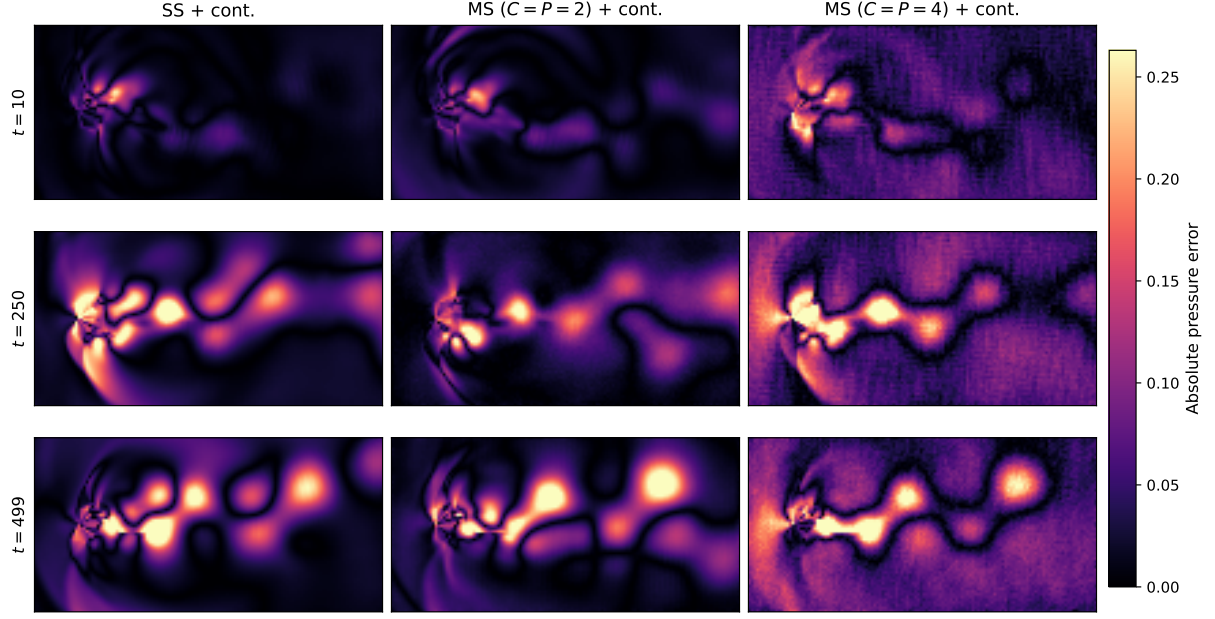


Figure 4: Absolute pressure-error maps for the Tra case at rollout indices $t = 10, 250$, and 499 . Columns show the continuity-regularised single-step model and the $C = P = 2$ and $C = P = 4$ multi-step models. The initial rollout step is omitted because its errors are close to zero and provide little spatial contrast.

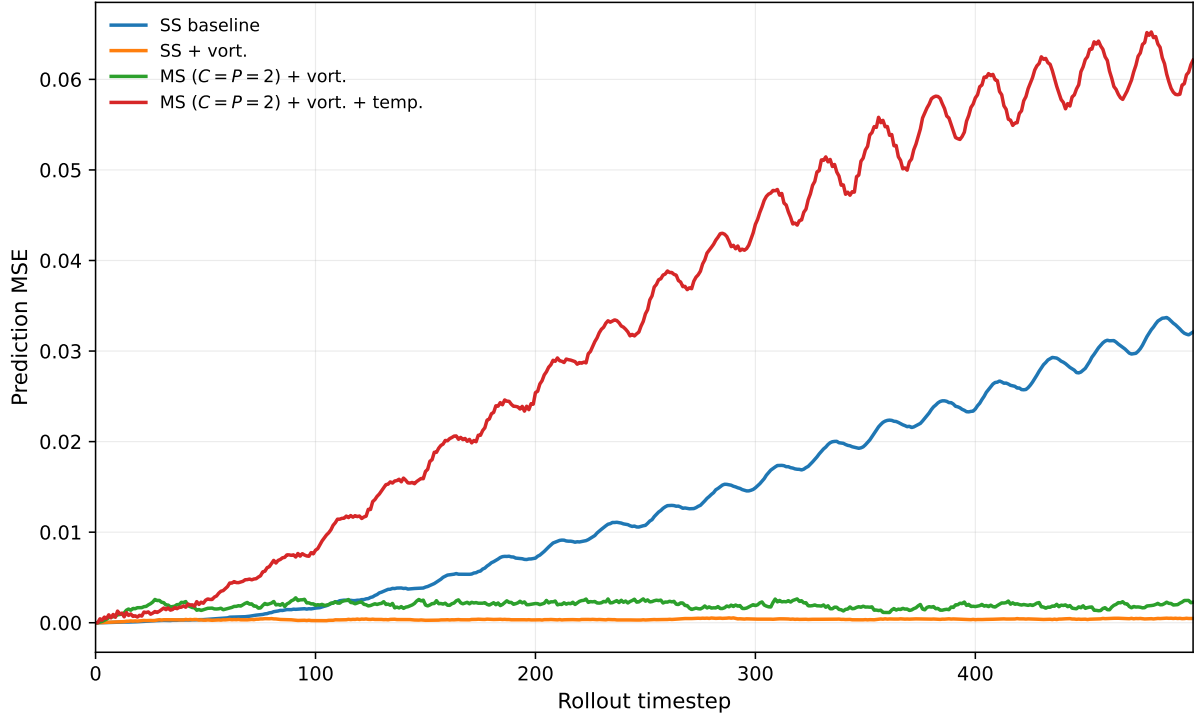


Figure 5: Prediction-MSE evolution over a 500-step rollout for representative Inc models in the high-Reynolds-number regime. The vorticity-regularised single-step model maintains the lowest error throughout the rollout, while the basic $C = P = 2$ multi-step model remains below the diffusion-only baseline but above the vorticity-regularised single-step model. The additional temporal-loss variant exhibits substantially larger error growth.



Figure 6: Prediction-MSE evolution over a 500-step rollout for the continuity-regularised Tra single-step and multi-step models. The multi-step variants generally maintain lower prediction errors than the single-step reference, while the differences among $C = P = 2, 3$, and 4 remain comparatively small over much of the rollout.

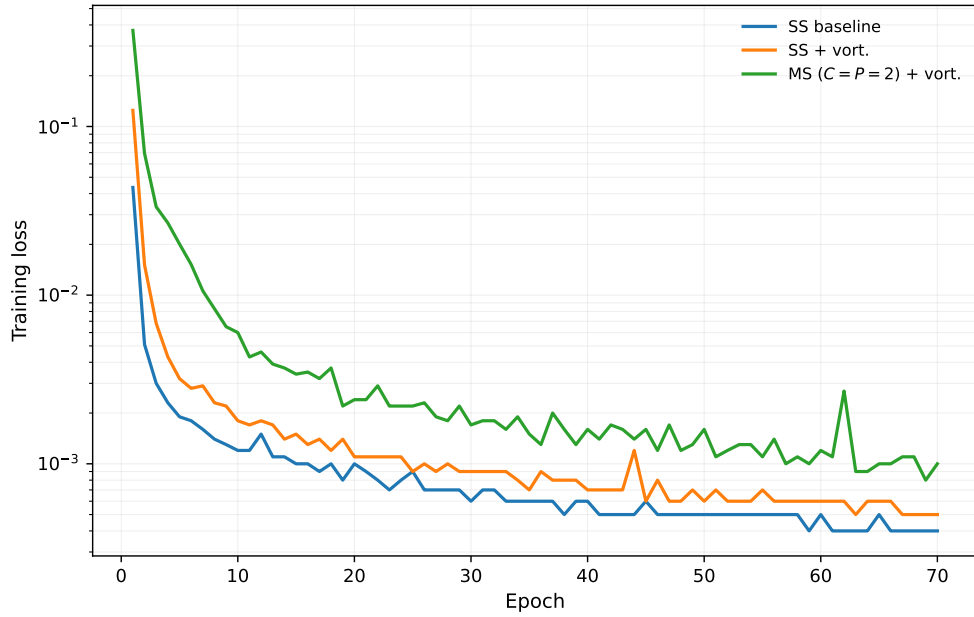


Figure 7: Representative training-loss curves for the Inc models. All three configurations show decreasing training loss over 70 epochs, while the $C = P = 2$ multi-step model converges to a higher training-loss level than the single-step models.

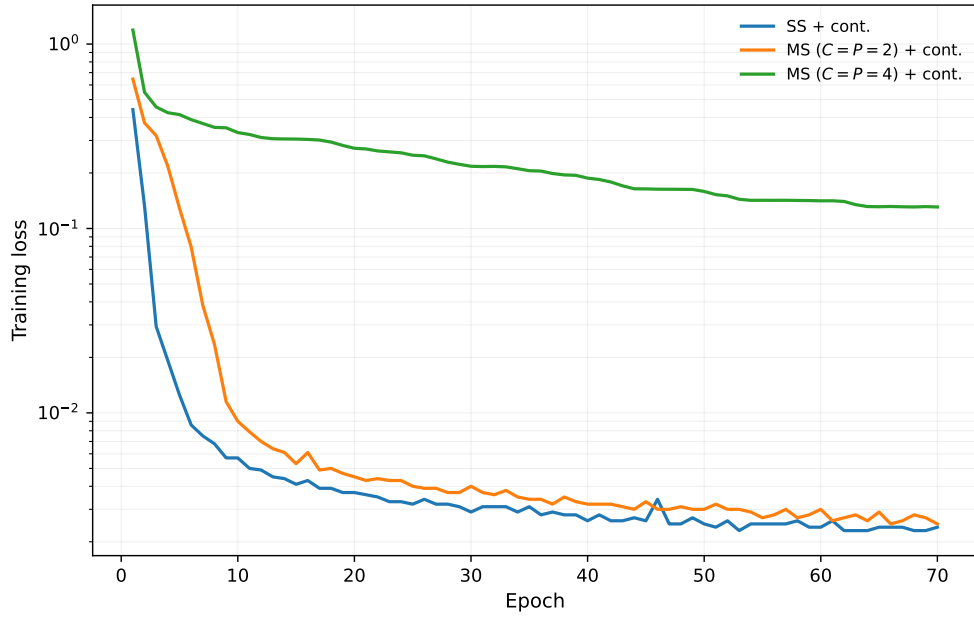


Figure 8: Representative training-loss curves for the continuity-regularised Tra models. The single-step and $C = P = 2$ models converge to similar low training-loss levels, while the $C = P = 4$ model retains a higher loss throughout training.

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